

PiXL KnowIT!

GCSE Physics

AQA Topic – Atomic structure

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Atoms and isotopes

- Structure of an atom
- Mass number, atomic number and isotopes
- Development of the atomic model

Atoms and nuclear radiation

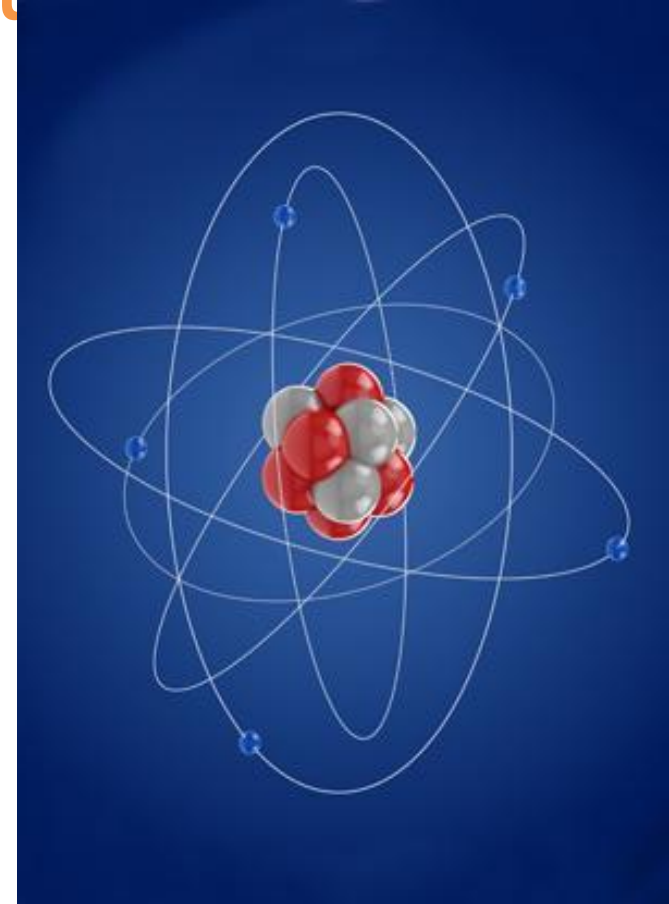
- Radioactive decay and nuclear radiation
- Nuclear equations
- Half lives and random nature of decay
- Radioactive contamination

Hazards and uses of radioactive emissions (physics only)

- Background radiation
- Different half lives of radioactive isotopes
- Uses of nuclear radiation

Nuclear fusion and fission (physics only HT)

- Nuclear fission
- Nuclear fusion



Atoms and isotopes

- [illegible]

Atoms are made up of different numbers of protons, neutrons and electrons. Atoms have the same number of + protons as – electrons so they are electrically neutral.

Electrons (-1 charge) arranged in **orbits or energy levels** around the nucleus.

Energy levels can hold a maximum of:

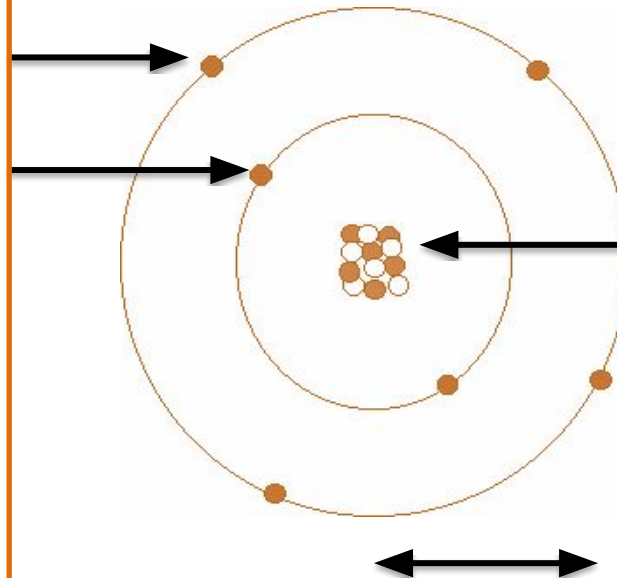
2 e⁻ in the first level

8 e⁻ in the second level

8 e⁻ in the third level

The radius of the nucleus is less than **1/10 000** the radius of the atom – the atom is **99.9999999%** empty space!

An atom of Carbon



← 9×10^{13} atoms in this dot of ink

Nucleus made up of:

Protons: charge +1

Neutrons: charge 0

Nucleus of an atom has a positive charge

Radius of an atom $1 \times 10^{-10}\text{m}$

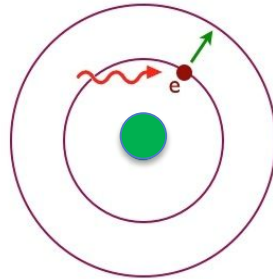
The nucleus holds **99%** of the mass of the atom

Electrons can absorb **electromagnetic radiation**. This excites the electron and can cause it to “jump” to a **higher energy level**. It can then release this energy as an electromagnetic wave by falling back to its original energy level.

Electromagnetic radiation

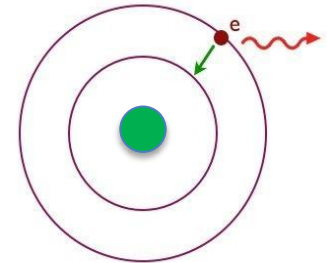


absorbed by the electron causes it to move to a higher energy level.



The electron can emit this stored energy as electromagnetic radiation.

As it loses energy the electron returns to its original energy level.



A photon is the amount of energy needed to make an electron jump an energy level. This same amount of energy will be emitted as a photon of electromagnetic radiation as the electron drops back to its original ground state.

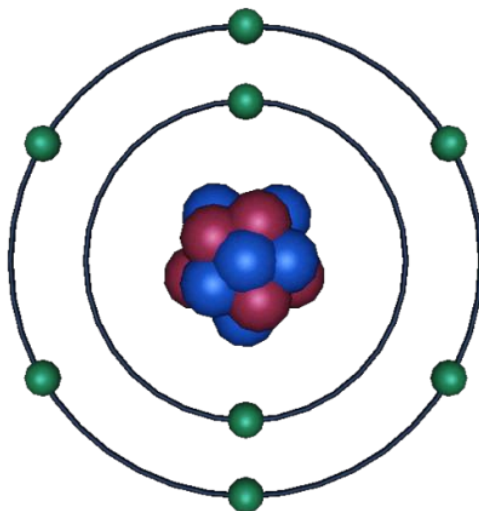
All atoms of a particular element have the **same number of protons**.
The number of protons in an element is called its **atomic number**.

7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00
15 P Phosphorus 30.97	16 S Sulfur 32.06	17 Cl Chlorine 35.45

Protons

On the periodic table, **oxygen** is shown as having an **atomic number** of eight, therefore **8 protons**.

- Protons
- Neutrons
- Electrons



Neutrons

The total number of **protons and neutrons** in an atom is called its **mass number**.
Oxygen has a mass number of 16. If it has 8 protons it must therefore have **8 neutrons** to make a mass number of 16.

Electrons

Atoms are electrically neutral so there must be the **same number** of electrons (-) as protons (+); **8 electrons**.

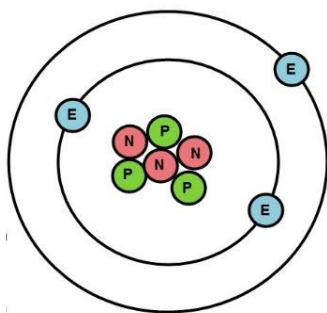
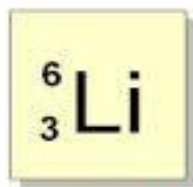
Oxygen has: 8 protons, $(16 - 8) = 8$ neutrons, and 8 electrons

Isotopes are elements with **different atomic masses**.

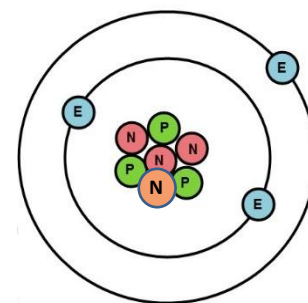
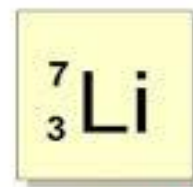
The number of protons can not change or it would not be the same element so **an isotope is an element with different numbers of neutrons**.

Lithium has two stable isotopes, Lithium 6 and Lithium 7

Lithium 6 has
3 protons
3 neutrons



Lithium 7 has
3 protons
4 neutrons

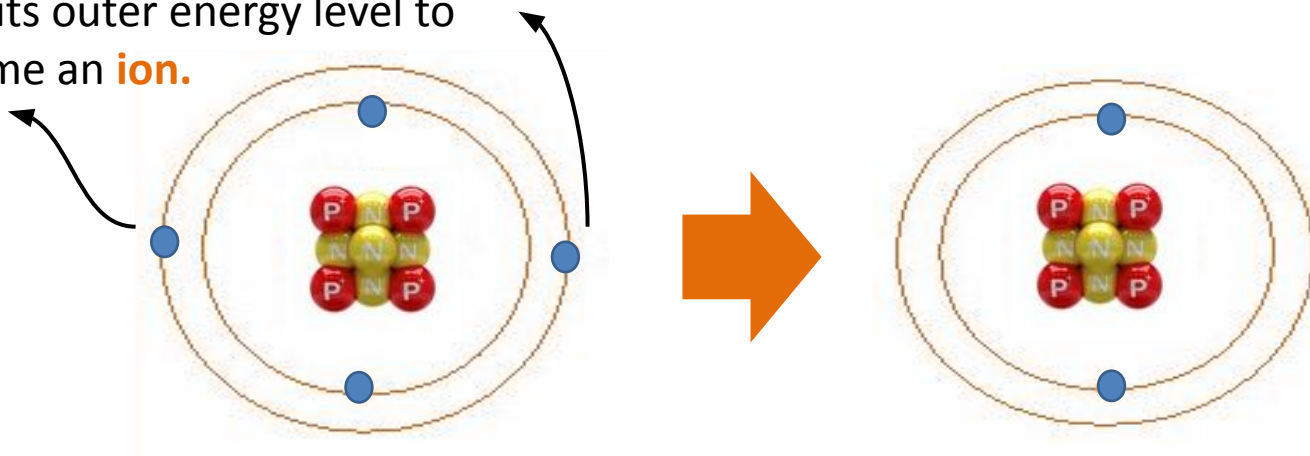


Both isotopes have the **same number of protons** and the **same number of electrons**.

Only the number of **neutrons changes** in an isotope.

Atoms can form **ions** if they gain or lose **electrons**. Atoms do this so they have **full outer energy levels**.

Beryllium **can lose 2 electrons** from its outer energy level to become an **ion**.



If Beryllium
loses 2 e⁻
it now has:

4 protons	4+
2 electrons	2-
	<u>2+</u>

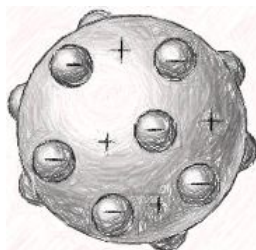
Beryllium ²⁺

Atoms can **lose (-) electrons** to become **positive (+) ions**
or **gain (-) electrons** to become **negative (-) ions**.

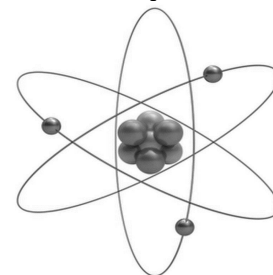
Pre 1900



Pre 1911



1911 to present



Sphere

**Plum pudding
model**

Nuclear model

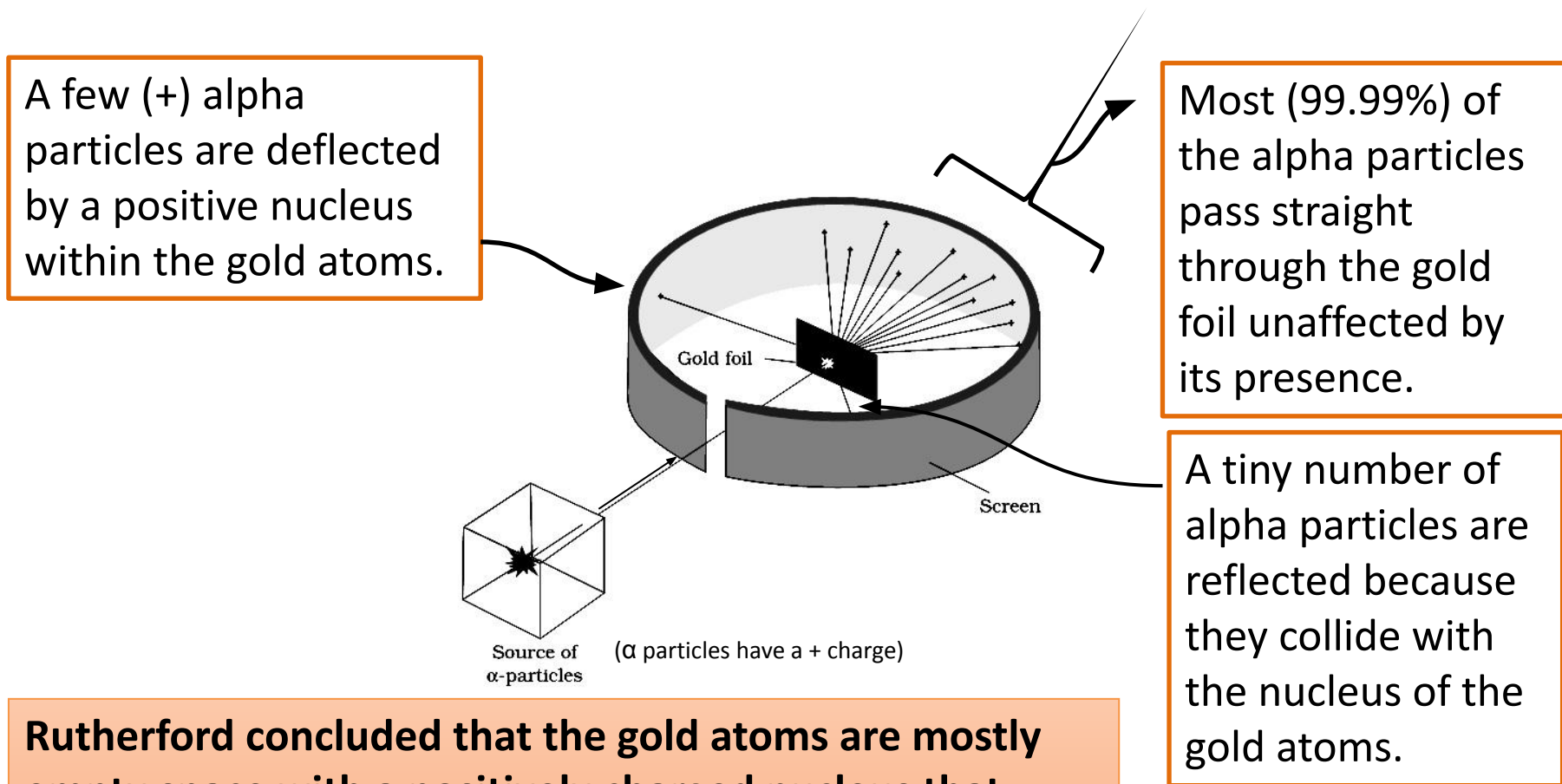
Before the discovery of the electron, atoms were thought to be **tiny spheres that could not be divided**.

The **discovery of the electron** led to the **plum pudding model** of the atom. The **plum pudding model** suggested the **atom is a ball of positive charge with negative electrons embedded in it**.

- **Alpha scattering experiment** – mass of the atom is concentrated in the **nucleus, which is charged**.
- **Niels Bohr** – **electrons orbit nucleus at different distances**.
- **Later experiments** – **positive charge in nucleus** divided into whole number of **smaller particles with positive charge**.
- **James Chadwick** – 20 years after nucleus accepted – provided evidence for existence of **neutrons in nucleus**.

Rutherford's alpha scattering experiment

A beam of alpha particles are directed at a very thin gold foil screen.



Rutherford concluded that the gold atoms are mostly empty space with a positively charged nucleus that contains nearly all the mass of the atom.

QuestionIT

Atoms and isotopes

- Structure of an atom
- Mass number, atomic number and isotopes
- Development of the model of the atom.



1. The diameter of an atom is about 0.000 000 000 2 m. What is this distance in standard form?
2. What is the nucleus of an atom composed of?
3. Describe what happens when an electron drops to a lower energy level in an atom.
4. An atom of sodium is represented by: $^{23}_{11}\text{Na}$

Use this information to determine the number of protons, neutrons and electrons in an atom of sodium.

5. What is the electrical charge attached to:
 - a neutron
 - an electron
 - a proton

1. The diameter of an atom is about 0.000 000 000 2 m. What is this distance in standard form?

$2 \times 10^{-10} \text{ m}$

2. What is the nucleus of an atom composed of?

Protons and neutrons (except Hydrogen which has no neutrons).

3. Describe what happens when an electron drops to a lower energy level in an atom.

It releases a photon of electromagnetic radiation.

4. An atom of sodium is represented by: $^{23}_{11}\text{Na}$

5. Use this information to determine the number of protons, neutrons and electrons in an atom of sodium.

Protons = 11 Neutrons = 12 Electrons = 11

6. What is the electrical charge attached to:

a neutron **Neutral**

an electron **Negative**

a proton **Positive**

6. What is the mass number and the atomic number for fluorine?



7. Beryllium has the chemical symbol:



Use this information to draw a representation of an atom of beryllium.

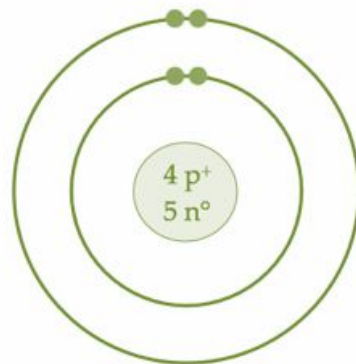
8. A different isotope of beryllium has an extra neutron. Give the chemical symbol of this new isotope of beryllium.

6. What is the mass number and the atomic number for fluorine?

Mass number = 19; atomic number = 9

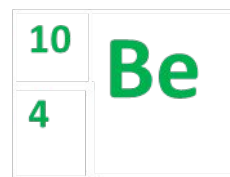


7. Beryllium has the chemical symbol:

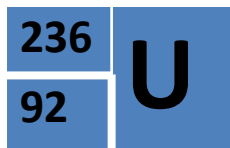


Use this information to draw a representation of an atom of beryllium.

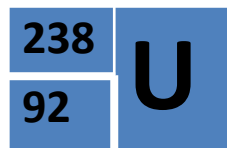
8. A different isotope of beryllium has an extra neutron. Give the chemical symbol of this new isotope of beryllium.



9. The radioactive element Uranium has two common isotopes.



and

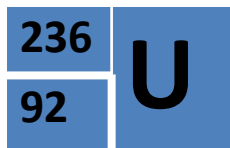


Complete the table to show the number of protons, neutrons and electrons in each isotope.

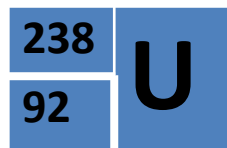
Isotope	Protons	Neutrons	Electrons
$\begin{array}{ c c } \hline 236 & \\ \hline 92 & \text{U} \\ \hline \end{array}$			
$\begin{array}{ c c } \hline 238 & \\ \hline 92 & \text{U} \\ \hline \end{array}$			

10. Sodium can lose its outer electron to have a full outer energy level. What will the atom now become?

9. The radioactive element Uranium has two common isotopes.



and



Complete the table to show the number of protons, neutrons and electrons in each isotope.

Isotope	Protons	Neutrons	Electrons

10. Sodium can lose its outer electron to have a full outer energy level. What will the atom now become?

An ion with a charge of 1+

11. Which scientific discovery resulted in the solid atom theory being adapted into the “*plum pudding*” model of the atom?
12. Rutherford carried out an experiment to show alpha particles either passing through gold leaf, being scattered by it. Summarise the conclusions he made from this experiment.
13. What contribution did Niels Bohr make to the arrangement of electrons in the atomic model?

11. Which scientific discovery resulted in the solid atom theory being adapted into the “*plum pudding*” model of the atom?

Discovery of the electron (in 1897).

12. Rutherford carried out an experiment to show alpha particles either passing through gold leaf, being scattered off it. Summarise the conclusions he made from this experiment.

Most passing through suggests atoms are mostly empty space.

Some being deflected suggests the nucleus has the same charge as the alpha particle (positive).

A few reflected suggests the nucleus is where most of the mass of the atom is.

13. What contribution did Niels Bohr make to the arrangement of electrons in the atomic model?

He realised that electrons orbit the nucleus in clearly defined energy levels, at different distances from the nucleus.

AnswerIT!

Atoms and isotopes

- Structure of an atom
- Mass number, atomic number and isotopes
- Development of the model of the atom.

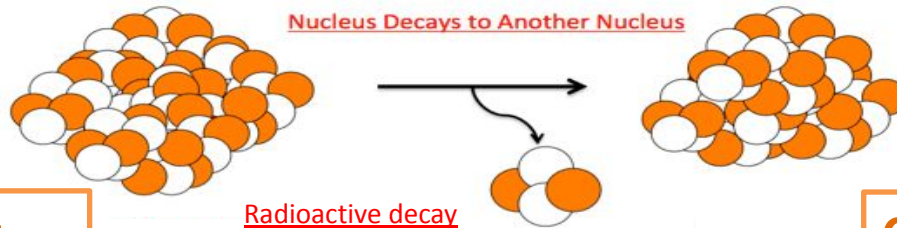


Atoms and nuclear radiation

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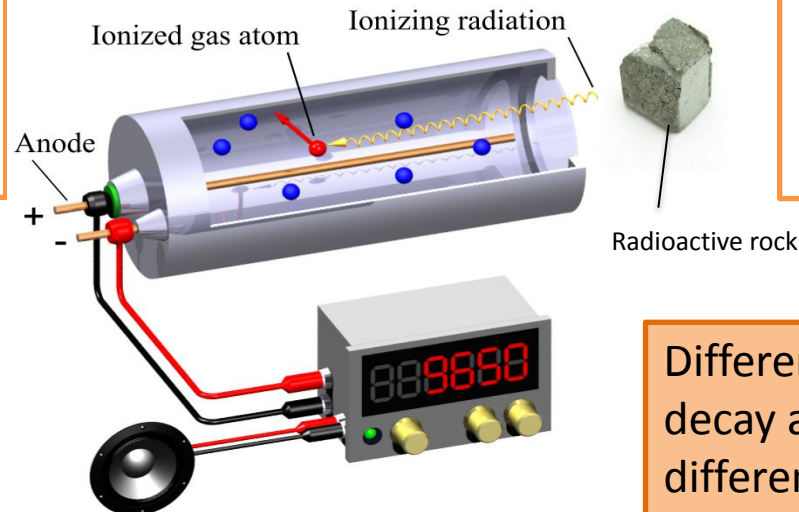
The nuclei of some atoms are unstable.
To become more stable these nuclei give out radiation.
This process is called radioactive decay.

Unstable
atom



Stable
atom

Activity = rate at which a source of unstable nuclei decays, measured in **becquerels (Bq)**.



Count-rate = number of decays recorded each second by a **detector** (e.g. Geiger-Muller tube)

Different radioactive isotopes decay at different rates and emit different types of radiation.

There are three types of radioactive decay, **alpha**, **beta** and **gamma**. All come from the **nucleus of the atom**. In the examples below, only the nucleus is shown.



Alpha (symbol α or ${}^4_2\text{He}$) consist of **2 protons and 2 neutrons** emitted from the nucleus. They have a **positive** charge as they contain 2 (+) protons.

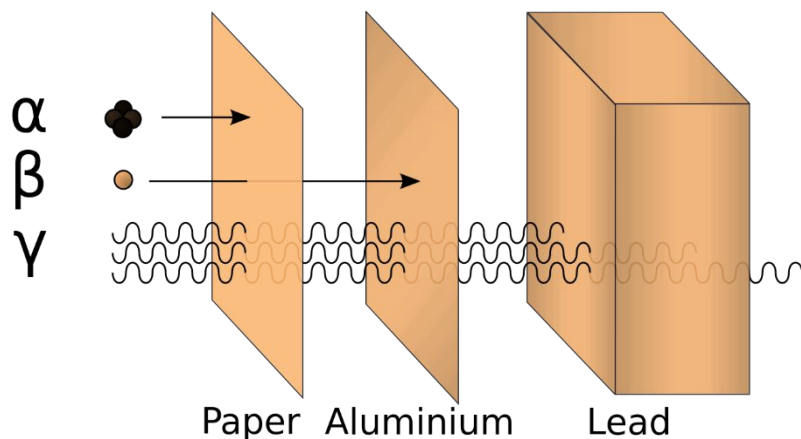


Beta (symbol β or ${}^0_{-1}e$) consist of an **electron** emitted from the nucleus. This results from a neutron splitting into a proton and an electron. Beta particles are **negatively** charged.



Gamma rays (symbol γ) are **electromagnetic radiation** emitted from the nucleus. Gamma radiation has **no mass** and **no electrical charge**.

Properties of alpha, beta and gamma radiation.



Alpha, beta and gamma radiation can penetrate different materials due to their differing nature.

Alpha – easily stopped by **a few sheets of paper**.

Beta – penetrates paper but stopped by a thin **sheet of aluminium**.

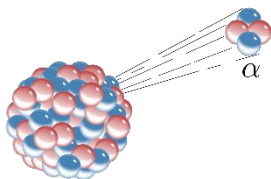
Gamma – only stopped by **thick lead** or several metres of **concrete**.

All three types of radiation cause **ionisation** of other atoms. If these atoms are in **living cells** it can cause damage which could lead to **cancer**.

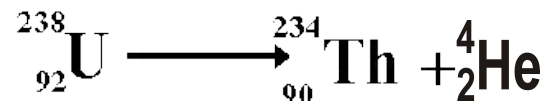
Name	Symbol	Speed	Range in air	Ionising power
Alpha	α	Slowest	6 - 8 cm	High
Beta	β	Medium	1 – 2 m	Medium
Gamma	γ	Fastest	300 -500 m	Low

Nuclear equations show the changes to an atom when it emits radiation.

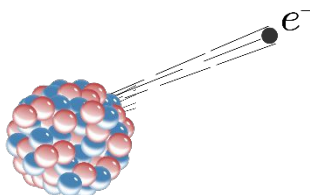
Alpha
emission



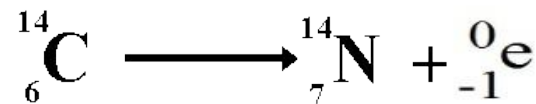
Nucleus **loses 2 protons** and **2 neutrons**.
Atomic number will reduce by 2 and atomic mass by 4.



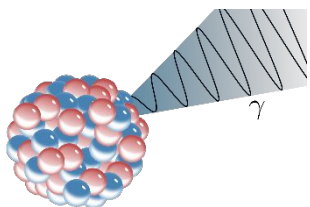
Beta emission



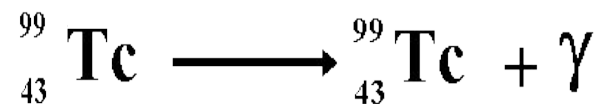
Nucleus **loses an electron** which is produced when a neutron turns into a proton. So **mass stays the same** but **atomic number of the product increases by one.**



Gamma
emission



No particles are emitted so there is **no change to the nucleus**. Atomic mass and atomic number stay the same.



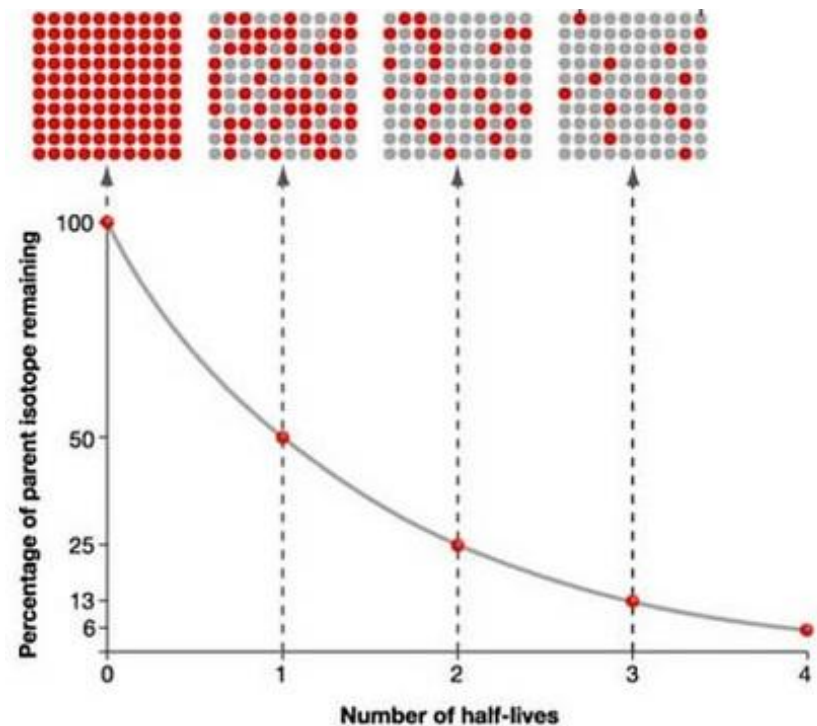
Radioactive decay is a random process so the likelihood of a decay taking place is a probability problem. For this reason, the **half-life** of an isotope is given rather than saying how long it will take to fully decay.

The **half-life** of a radioactive isotope is the time it takes for the **number of nuclei** of the isotope in the sample to halve, or the time it takes for the **count rate** from a sample containing the isotope to fall to half its initial level.

The net decline of the isotope is the fraction remaining after a number of half lives.

E.g. $100 \rightarrow 50 \rightarrow 25$

After 2 half lives net decline is $75/100 = \frac{3}{4}$

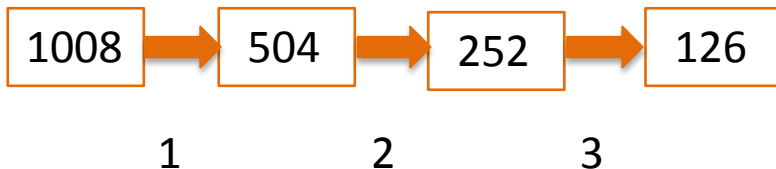


Calculating the half life of a radioactive isotope.

If you know the start and finish count rate and the time taken, you can calculate the half life.

Example:

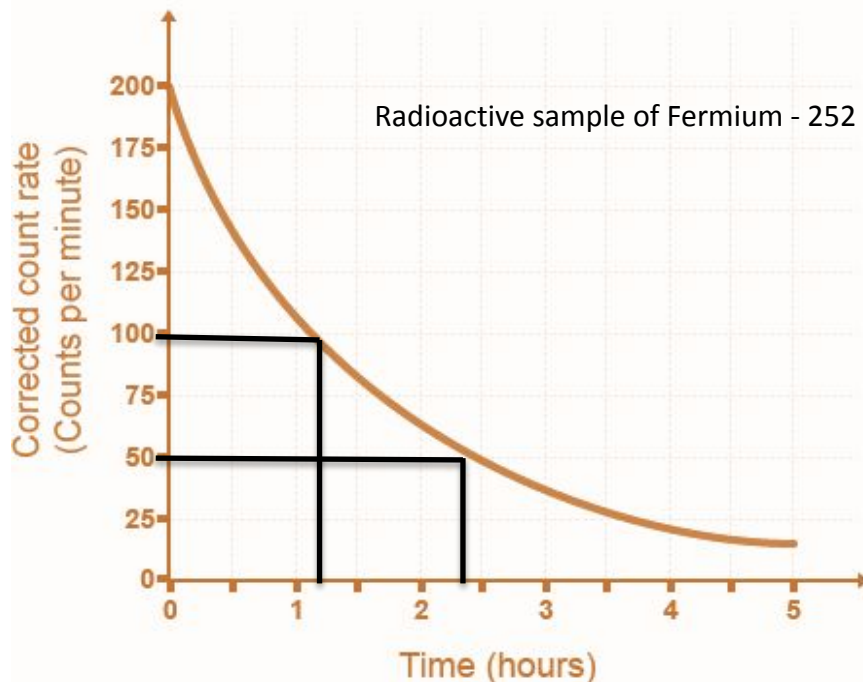
The count rate of an isotope is 1008 Bq. This falls to a count rate of 126 over a period of 21 days.



3 half lives for count rate to fall to 126.

These 3 half lives took 21 days so each half life took 7 days.

Half life if this isotope = 7 days



200 counts / min at the beginning.

100 counts/min occurred after 1.2 hours.

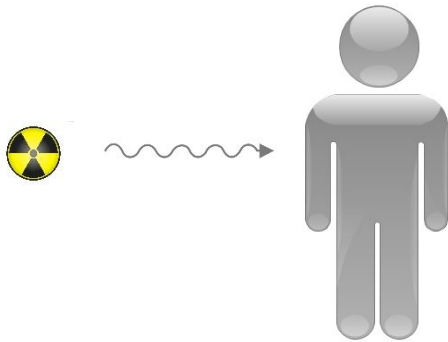
50 counts/min occurred after 2.4 hours.

It always takes 1.2 hours for the count rate to halve.

Half life of Fermium - 252 = 1.2 hours.

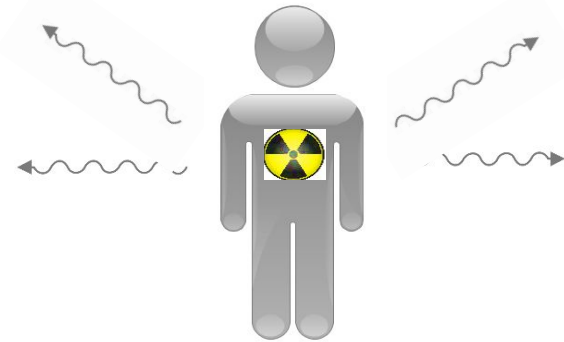
Radioactive substances can be hazardous by contamination or irradiation.

Irradiation is when an object or person is **exposed** to radiation. Protection from irradiation means stopping the radiation from reaching you.



Medical dressings are often irradiated but present no danger to the user.

Contamination is when a radioactive source is in **contact** with an object or person. The radioactive substance rather than the emissions are present.



The object remains radioactive until the contamination is removed or decays naturally.

QuestionIT

Atoms and nuclear radiation

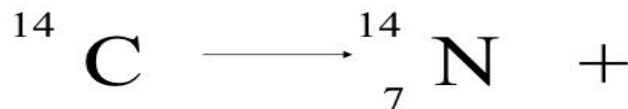
- Radioactive decay and nuclear radiation
- Nuclear equations
- Half life and the random nature of radioactive decay
- Radioactive contamination



1. Which part of an atom is involved with radioactive decay?
2. Explain the meaning of the term activity, as applied to radioactive materials and state the units of activity.
3. What is meant by the term “count rate”?
4. Copy and complete the table to show the nature of alpha, beta and gamma radiations.

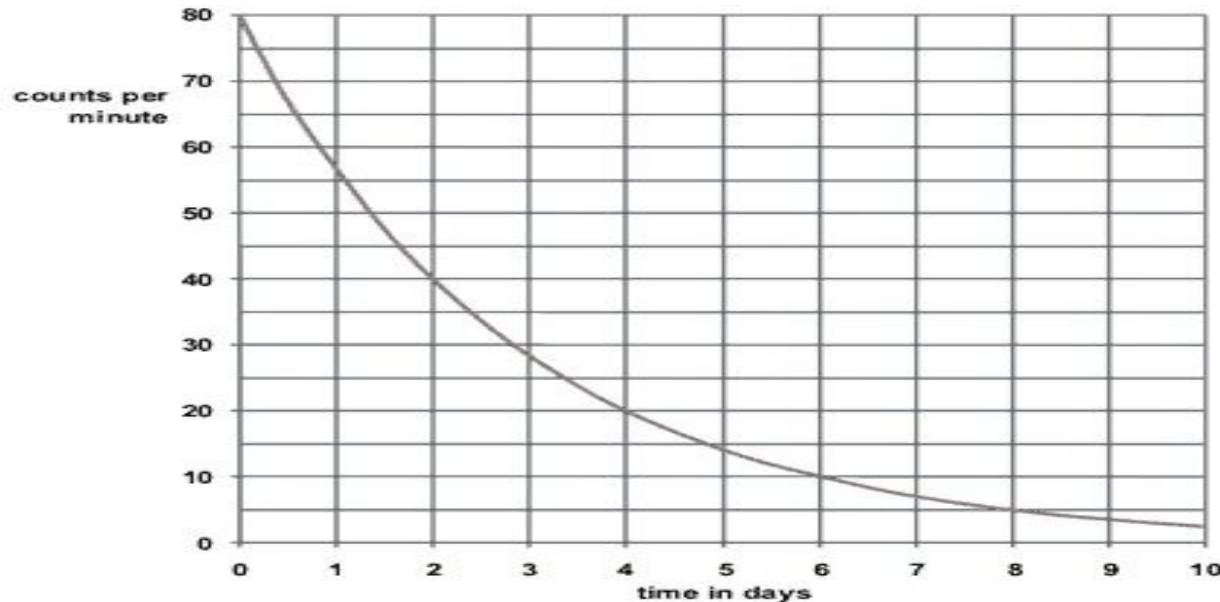
Radiation	Symbol	Composition	Electrical charge
Beta	β		
Gamma		Electromagnetic wave	
Alpha			+2

5. A piece of radioactive rock shows a reading of 350 counts/min. When covered in aluminium foil, this drops down to 4 counts/min. Explain which type of radiation this rock is emitting.
6. Radioactive emissions are often described as ionising radiations. What does this mean?
7. Smoke detectors use americium-241 which is an alpha emitter. Explain why an alpha source is used in these detectors.
8. Why is an alpha particle often described as a helium nuclei?
9. Complete the nuclear equation for the beta decay of carbon.



10. Uranium-235 undergoes an alpha decay to produce thorium-231. (atomic number of uranium is 92). Complete the nuclear equation for this process.
11. When iodine 131 decays, there is no mass change in the nucleus and no new products formed. What type of radioactive emission is this?
12. Explain what is meant by the term “half life”.
13. A radioactive sample reduces its count rate from 240 counts/min to 30 counts/min over a period of 60 hours what is its half life?

14. Use the decay curve below to work out the half-life of the isotope.



15. (Physics only) Calculate the net decline of the above isotope expressed as a ratio, during radioactive emission after 3 half-lives.

16. Explain the difference between radioactive irradiation and radioactive contamination.

17. Copy and complete the table below to suggest one way of preventing exposure to irradiation and contamination by radioactive materials.

Type of exposure	Method of preventing exposure
Irradiation	
Contamination	

AnswerIT!

Atoms and nuclear radiation

- Radioactive decay and nuclear radiation
- Nuclear equations
- Half life and the random nature of radioactive decay
- Radioactive contamination



1. Which part of an atom is involved with radioactive decay?

The nucleus only.

2. Explain the meaning of the term activity as applied to radioactive materials and state the units of activity.

The rate at which a source of unstable nuclei decays. Units Bq.

3. What is meant by the term “count rate”?

The number of radioactive decays recorded in a given time.

4. Copy and complete the table to show the nature of alpha, beta and gamma radiations.

Radiation	Symbol	Composition	Electrical charge
Beta	β	an electron	-1
Gamma	γ	Electromagnetic wave	0
Alpha	α	2 protons and 2 neutrons	+2

5. A piece of radioactive rock shows a reading of 350 counts/min. When covered in aluminium foil, this drops down to 4 counts/min. Explain which type of radiation this rock is emitting.



Could be alpha or beta as both would be stopped by the foil and gamma would not be stopped by the foil.

6. Radioactive emissions are often described as ionising radiations. What does this mean?

The emissions knock off electrons from atoms which then become ions.

7. Smoke detectors use americium-241 which is an alpha emitter. Explain why an alpha source is used in these detectors.

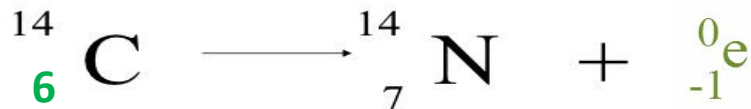
Alpha particles are easily stopped by smoke.

They do not travel far in air so are safe for the user.

8. Why is an alpha particle often described as a helium nuclei?

It contains 2 protons and 2 neutrons, the same as the nucleus of a helium atom.

9. Complete the nuclear equation for the beta decay of Carbon.



10. Uranium-235 undergoes an alpha decay to produce thorium-231. (atomic number of Uranium is 92). Complete the nuclear equation for this process.



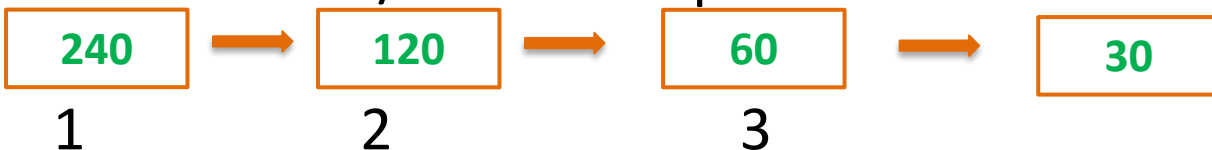
11. When iodine 131 decays, there is no mass change in the nucleus and no new products formed. What type of radioactive emission is this?

Gamma emission

12. Explain what is meant by the term “half life”.

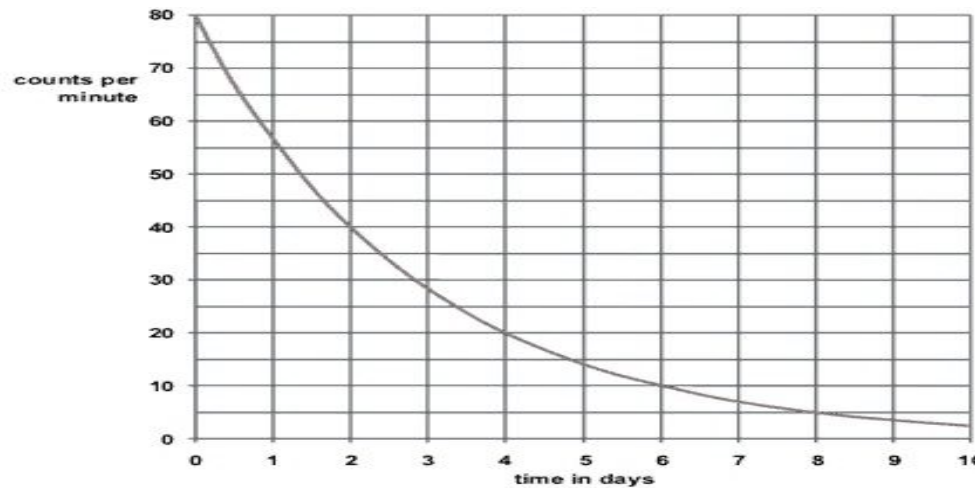
The time it takes a radioactive sample to lose half its radioactivity (as measured by count rate).

13. A radioactive sample reduces its count rate from 240 counts/min to 30 counts/min over a period of 60 hours what is its half life?



Three half lives in 60 hours = 20 hour half life

14. Use the decay curve below to work out the half-life of the isotope.



80 = 0 day; 40 = 2 days. Difference = 2 days. Half-life = 2 days

15. (Physics only) Calculate the net decline of the above isotope expressed as a ratio, during radioactive emission after 3 half-lives.

Counts/ min reduce from 80 to 10 in 3 half-lives.

Decline is 70/80 or 7/8ths

16. Explain the difference between radioactive irradiation and radioactive contamination.

Irradiation is exposure to emissions from radioactive materials that are not in contact with an object. Contamination is when radioactive materials are in contact with the object.

17. Copy and complete the table below to suggest one way of preventing exposure to irradiation and contamination by radioactive materials.

Type of exposure	Method of preventing exposure
Irradiation	Wear protective clothing, e.g., lead apron, to shield from radiation. Move away from the radiation. Shield the radiation with appropriate material.
Contamination	Avoid contact with radioactive materials. Prevent radioactive materials being released into the environment.

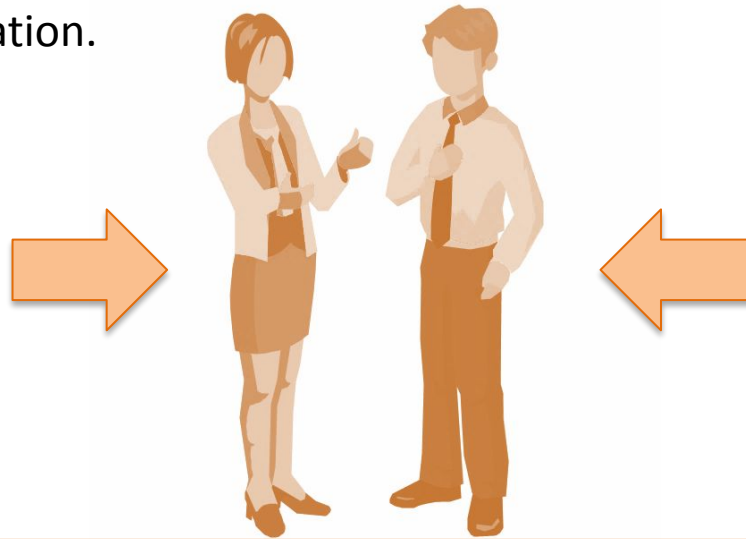
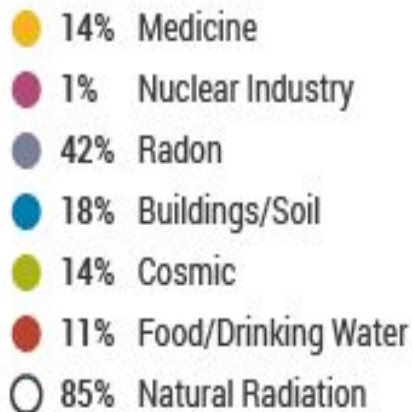
Hazards and uses of radioactive emissions and of background radiation (Physics only)

- [illegible]

Background radiation is the constant , low level radiation in the environment. This can be natural radiation from rocks, building materials, cosmic rays etc.

Radioactive pollution from nuclear testing, nuclear power and industrial/medical waste also contributes to background radiation.

Sources of radioactive exposure and contamination.



Radiation dose is measured in:
sieverts (Sv)

1 Sv = 1000
millisieverts

Everyone receives background radiation but people who **work or live** in locations with high levels of radiation **receive additional doses of radiation**.

Some nuclear workers, medical staff, military and industrial workers may have higher doses due to working with radioactive sources.

Radioactive isotopes have an enormous range of half-lives.

Examples of the range of half-lives of radioactive materials

Radioactive nuclide	Nuclide notation	Half-life
Lithium-8	${}^8_3\text{Li}$	0.838 s
Krypton-89	${}^{89}_{36}\text{Kr}$	3.16 minutes
Sodium-24	${}^{24}_{11}\text{Na}$	15 hours
Iodine-131	${}^{131}_{53}\text{I}$	8 days
Cobalt-60	${}^{60}_{27}\text{Co}$	5.27 years
Radium-228	${}^{226}_{88}\text{Ra}$	1600 years
Uranium-235	${}^{235}_{92}\text{U}$	703 million years

Half-life and hazard

Radioactive isotopes with a **short half-life** often give **high doses** of radiation in a short period of time so are often dangerous.

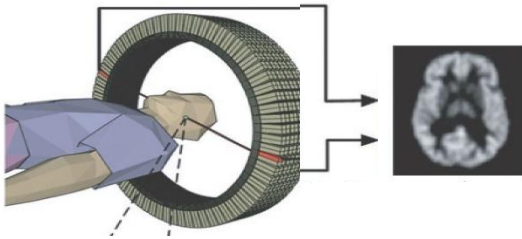
Long half-life isotopes are **low dose** hazards but they are around for a very **long time**.

Uranium-238 is the main fuel producer for the nuclear industry but is so slow at emitting radiation it is often considered quite safe by scientists.

Products of the nuclear industry such as Iodine-131 are much more dangerous as they emit radiation at a much faster rate and are soluble so they get into the food chain much more easily.

Nuclear radiations are used as **tracers** in the body to explore possible injury or disease of internal organs.

A **radioactive isotope** is either injected or ingested into the body, given time to circulate and **accumulate in damaged areas**. Then the emissions radiating out of the body are detected.

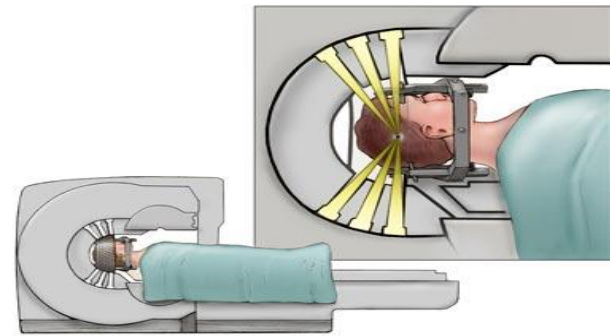


A camera such as a gamma detector or a PET scanner detects any accumulation of the tracer.

Tracers have to be **beta or gamma** emitters as alpha does not penetrate the body. The tracer must also have a **very short half-life** to minimise dosage.

Uses of nuclear radiation (medical)

Radiation therapy is used to treat illnesses such as cancer. **Cancer cells** are living cells and so are killed off by relatively high doses of **gamma rays**.



Here, the gamma rays are directed from the outside. The high dose required to kill the cancer cells will also kill healthy cells. The technique uses a 3 dimensional set of gamma ray guns all focussed on the cancer cells.

This **kills the cancer** cells while minimising the damage **healthy cells**.

QuestionIT

Hazards and uses of
radioactive emissions and
of background radiation
(Physics only)

- Background radiation
- Different half lives of radioactive isotopes
- Uses of nuclear radiation



1. Describe sources of background radiation, clearly identifying which are natural and which are man-made.
2. Describe **two** occupations where the radiation dose received by workers is likely to be higher than the background radiation.
3. Lithium-8 is a beta emitter with a half life of 0.8 s. What precautions would you take when working with this isotope?

1. Describe sources of background radiation, clearly identifying which are natural and which are man made.

Natural – rocks, cosmic radiation, building materials

Man made – fallout from nuclear weapons testing, nuclear power accidents, industrial and medical waste.

2. Describe **two** occupations where the radiation dose received by workers is likely to be higher than the background radiation.

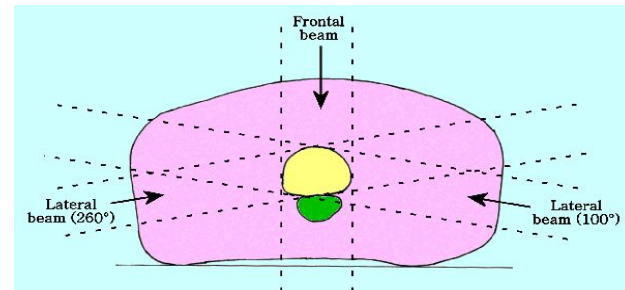
Working in the nuclear industry, as a radio-medical worker, working with radioisotopes in industry.

3. Lithium-8 is a beta emitter with a half life of 0.8 s. What precautions would you take when working with this isotope?

Take care not to ingest any material or come into contact with it (gloves, safety glasses, lab coat).

Isolate the material for 24 hours to allow it to decay to a safe level.

4. Radium - 226 is an alpha emitter with a half-life of 1600 years. Explain how the way this material is stored is influenced by these properties.
5. The diagram shows how three separate gamma beams are used to treat a cancer tumour. Why is this preferred to using one powerful beam?

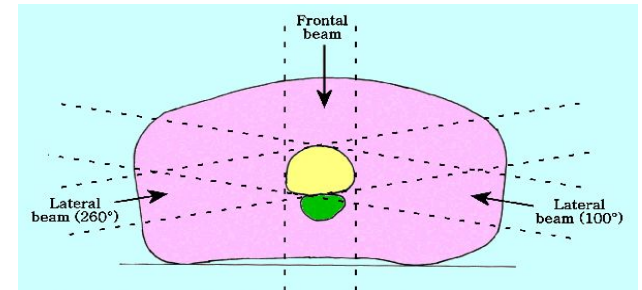


6. Alpha emitting radioisotopes cannot be used as tracers in the body to explore injured or diseased organs. Why?

4. Radium - 226 is an alpha emitter with a half-life of 1600 years. Explain how the way this material is stored is influenced by these properties.

Any sealed container will prevent radiation escaping as alpha particles are not very penetrating. Radioactive material will need to be placed in permanent storage, buried underground, as it will be radioactive for a very long time.

5. The diagram shows how three separate gamma beams are used to treat a cancer tumour. Why is this preferred to using one powerful beam?



Single beam will damage both healthy or cancer cells but all three beams are focussed on the tumour so these cells receive a triple dose of radiation to kill them and reduces harm to healthy cells.

6. Alpha emitting radioisotopes cannot be used as tracers in the body to explore injured or diseased organs. Why?

Alpha particles are highly ionising so they will cause damage to bodily cells. They are also easily absorbed by body tissue so they will not escape the body to be detected.

AnswerIT!

Hazards and uses of
radioactive emissions ar
of background radiation
(Physics only)

- Background radiation
- Different half lives of radioactive isotopes
- Uses of nuclear radiation



LearnIT! KnowIT!

Nuclear fusion and
fission (physics only –
Higher Tier)

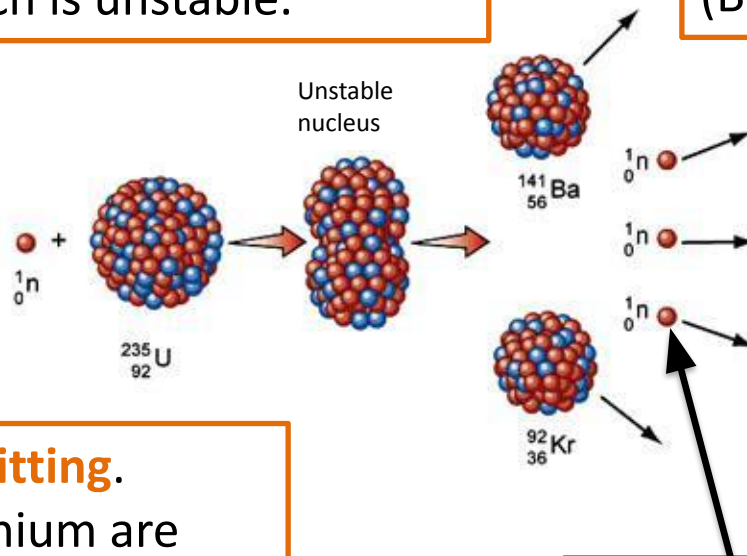
- Nuclear fission
- Nuclear fusion



Nuclear fission is the splitting of large, unstable atoms into two or more atoms along with the release of energy.

Here, a nucleus of Uranium - 235 is **bombarded with a neutron** to form Uranium 236 which is unstable.

Two **smaller elements** are produced in this fission process (Barium and Krypton).



ENERGY

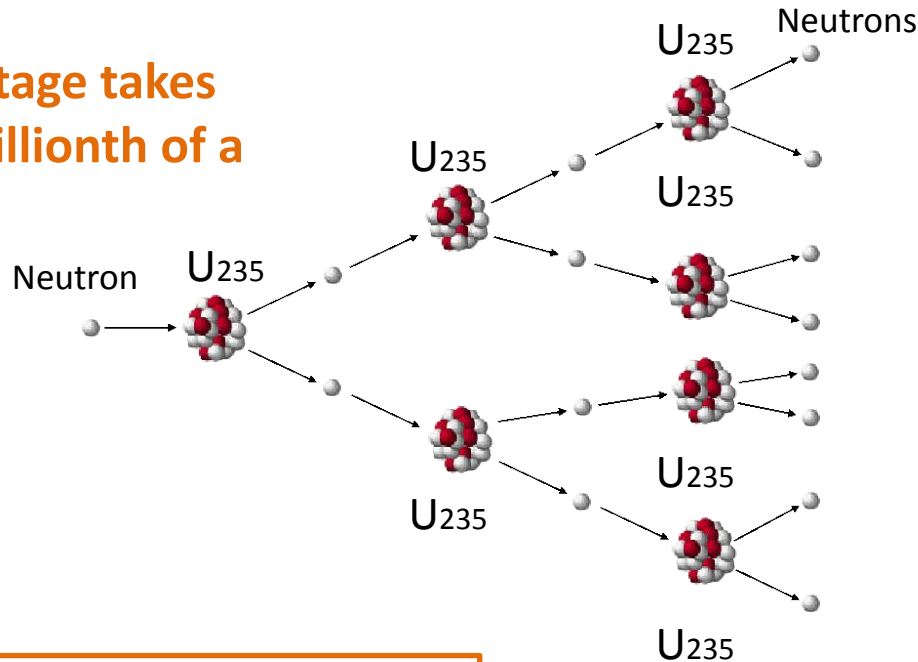
All the fission products have **kinetic energy**

Fission means **splitting**. Uranium or Plutonium are often used in nuclear reactors to produce heat as their nuclei are easy to split.

These **neutrons** may go on to **split further Uranium atoms**.

Nuclear **fission releases more neutrons** which can lead to further fission reactions. If this is uncontrolled a **chain reaction** can occur which will release **vast amounts of energy**.

Each fission stage takes less than a millionth of a second!

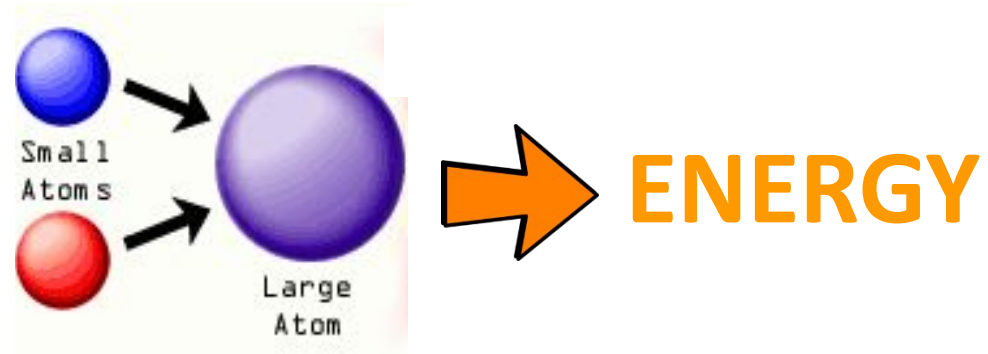


If the chain reaction is **not controlled** a vast amount of energy is released almost instantly. This is how a **nuclear weapon** works.

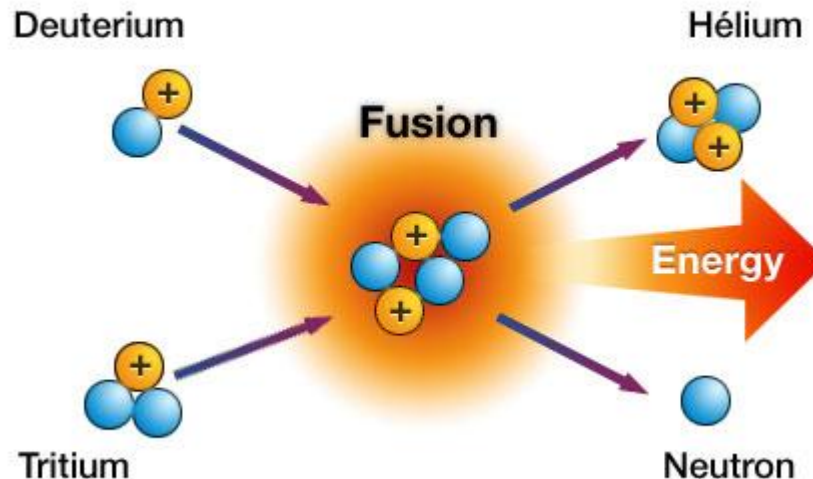
In a **nuclear reactor**, some of the **neutrons are absorbed** by boron rods to **control the reaction** and hence control the amount of energy released.

Nuclear **fusion** is the **joining** of two small (light) nuclei to form a larger nucleus.

When two small nuclei join to form a larger nucleus, a small amount of **mass** is changed **into** a large amount of **energy**.



Fusion reactions take place in **stars** to release vast amounts of energy. Here, two types of hydrogen nuclei, deuterium and tritium, fuse to form helium and release a neutron plus energy.



QuestionIT!

Nuclear fusion and fission (physics only – Higher Tier)

- Nuclear fission
- Nuclear fusion



1. Which particle is needed to begin the fission of a large, unstable nuclei?
2. During the fission of uranium, two smaller nuclei are produced and what else?
3. Copy and complete the diagram below to show the chain reaction of a sample of uranium



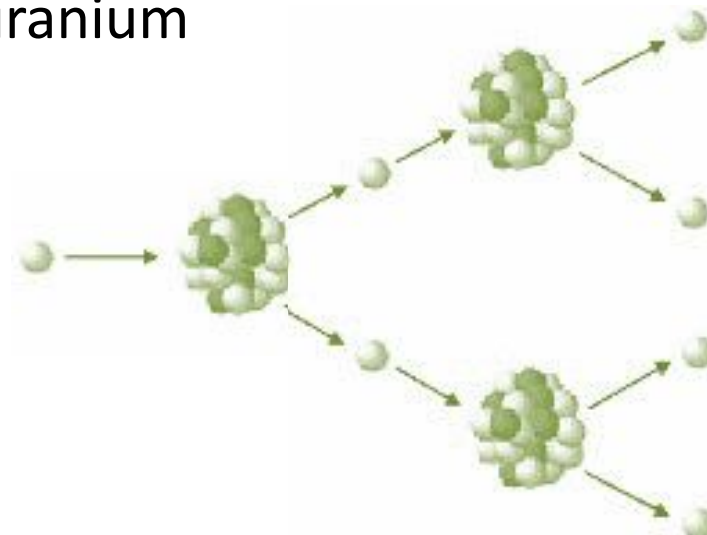
1. Which particle is needed to begin the fission of a large, unstable nuclei?

A neutron

2. During the fission of uranium, two smaller nuclei are produced and what else?

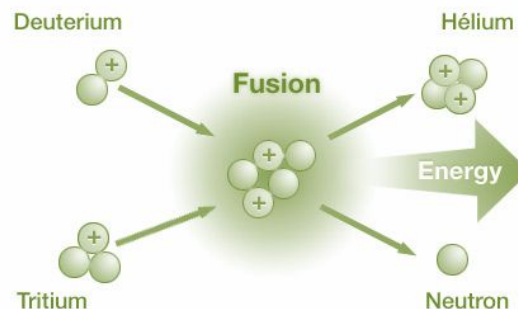
A number of neutrons and large amounts of energy.

3. Copy and complete the diagram below to show the chain reaction of a sample of uranium



4. Explain what is meant by a controlled chain reaction.
5. What is nuclear fusion?
6. Where does nuclear fusion take place on a large scale?
7. Draw a diagram to show the process of fusion between deuterium and tritium to produce energy.

4. Explain what is meant by a controlled chain reaction.
Nuclear fission reaction where some of the neutrons produced in the reaction are absorbed to prevent the reaction running out control.
5. What is nuclear fusion?
The joining of two small nuclei to form a single larger nucleus.
6. Where does nuclear fusion take place on a large scale?
In stars (sun)
7. Draw a diagram to show the process of fusion between deuterium and tritium to produce energy.



AnswerIT!

Nuclear fusion and fission (physics only – Higher Tier)

- Nuclear fission
- Nuclear fusion

